

Artificial Intelligence in DevOps Automation: A Systematic Literature Review of Applications, Benefits, Challenges, and Research Trends from 2016 to 2026

Abstract

Artificial Intelligence (AI) is increasingly being applied to DevOps to improve automation, monitoring, testing, incident management, operational decision support, and remediation. However, the literature remains fragmented across DevOps stages, AI techniques, evaluation settings, and levels of empirical maturity. This study presents a Systematic Literature Review (SLR) of Artificial Intelligence for DevOps Automation covering the period 2016–2026. The review followed systematic review guidance for software engineering, used PICOC to structure the review scope and search strategy, and applied PRISMA 2020 for study-selection reporting. Searches were conducted in ScienceDirect, IEEE Xplore, and ACM Digital Library, identifying 1,132 records. After removal, deduplication, screening, retrieval, and eligibility assessment, 60 primary studies were included and analyzed using a 26-field extraction schema, although substantial full-text retrieval limitations remained.

The results show that research activity became increasingly concentrated in 2024–2026 and evolved from predominantly predictive and diagnostic approaches toward Generative AI, Large Language Models, and emerging agentic systems. AI applications were concentrated primarily in monitoring and observability, operations, and incident/root-cause/remediation activities. Classical Machine Learning remained the dominant primary AI category, while Generative AI/LLM and Agentic/Multi-Agent approaches formed an emerging methodological segment. Reported benefits included improvements in speed and efficiency, detection quality, reliability, resource optimization, automation, security, and human decision support. However, the strength of evidence varied substantially across production, industrial empirical, controlled experimental, benchmark, simulation, and conceptual settings.

The review also identified recurring challenges related to data quality, generalizability, concept drift, evaluation realism, explainability, LLM reliability, cost, integration complexity, autonomous-remediation safety, human oversight, security, privacy, and evaluation inconsistency. These findings were synthesized into eight research gaps and five future research directions. Overall, the literature suggests a trajectory from Predictive → Diagnostic → Generative → Agentic → Trustworthy Autonomous DevOps, while indicating that technical capability is developing faster than demonstrated production benefit. Future progress therefore depends on closing the gap between technical feasibility and safe, reliable, explainable, cost-effective, and production-validated AI-enabled DevOps.

Keywords: Artificial Intelligence, DevOps, AIOps, Generative AI, Large Language Models, AI Agents, DevOps Automation, Systematic Literature Review

1. Introduction

DevOps has become a widely adopted software engineering approach for integrating development and operations activities through continuous integration, continuous delivery, automation, monitoring, and rapid feedback. Its adoption seeks to shorten delivery cycles while preserving software quality, reliability, and operational responsiveness [1]. As software systems become increasingly distributed, cloud-native, and continuously evolving, however, DevOps teams must process large volumes of operational telemetry, build and deployment information, incident reports, logs, traces, metrics, and configuration data.

Artificial Intelligence (AI) has increasingly been introduced into this operational environment to support or automate tasks that are difficult to perform manually at scale. Earlier work predominantly emphasized machine-learning-based prediction, anomaly detection, failure management, and operational analytics, often discussed under the broader AIOps paradigm [2]. More recent research extends this direction through Large Language Models (LLMs), which enable natural-language reasoning over incident data, logs, documentation, and operational knowledge [3], [4]. Emerging agentic approaches further combine LLM reasoning with tools, workflows, and

execution mechanisms, although these systems remain an early and safety-sensitive research direction rather than an established level of industrial maturity [5].

Despite the growing volume of research, the AI-for-DevOps literature remains heterogeneous. Studies differ substantially in the DevOps activities they address, the AI methods they employ, the datasets and operational environments used for evaluation, the benefits they report, and the strength of evidence supporting those benefits. Existing work also spans predictive analytics, diagnostic AIOps, generative approaches, and increasingly agentic systems, making it difficult to obtain a consolidated view of how the field has evolved and where reliable evidence currently exists.

This review therefore focuses specifically on AI-for-DevOps: research in which AI is used to improve, support, or automate DevOps activities. Studies concerned exclusively with MLOps or DevOps-for-AI are outside the review boundary. The review covers the period 2016–2026 and synthesizes evidence from ScienceDirect, IEEE Xplore, and ACM Digital Library using Kitchenham and Charters-style systematic review guidance, PICOC, and PRISMA 2020 reporting.

The search identified 1,132 records. After two non-article/container records were removed before screening and 317 duplicate occurrences were removed, 813 unique records were screened by title and abstract. A total of 382 reports were sought for retrieval; 73 full texts were obtained and assessed, 13 were excluded at full-text stage, and 60 primary studies were included in the final corpus.

This SLR makes five contributions. First, it characterizes research trends in AI-for-DevOps between 2016 and 2026. Second, it maps AI applications to DevOps stages and activities. Third, it develops a seven-category taxonomy of AI methods and technologies used across the corpus. Fourth, it synthesizes reported benefits while explicitly distinguishing the strength of supporting evidence. Fifth, it consolidates recurring challenges, research gaps, and future research directions toward trustworthy AI-enabled DevOps.

Accordingly, the review addresses five research questions: RQ1 examines research trends over time; RQ2 identifies the DevOps stages and activities where AI is applied; RQ3 analyzes the AI methods and technologies used; RQ4 synthesizes reported benefits and impacts; and RQ5 examines challenges, limitations, research gaps, and future opportunities.

Taken together, the review interprets the literature as moving from predictive and diagnostic AI toward generative and agentic approaches. This trajectory should not be interpreted as evidence that autonomous DevOps is already production-ready. Rather, it motivates a longer-term research direction toward trustworthy autonomous DevOps in which capability is matched by reliability, safety, evidence, governance, and human oversight.

The remainder of this paper is organized as follows. Section 2 presents the conceptual background and scope boundary. Section 3 describes the review methodology. Section 4 reports the results for RQ1–RQ5. Section 5 discusses the implications of those results. Section 6 addresses threats to validity, and Section 7 concludes the review.

2. Background

2.1 DevOps and DevOps Automation

DevOps integrates development and operations through collaborative practices, automation, continuous integration and delivery, monitoring, feedback, and shared responsibility for software delivery and operations [1]. Automation is central to this model because repeatable build, test, deployment, monitoring, and operational workflows reduce manual effort and enable faster feedback.

In this review, a distinction is made between procedural automation and intelligent automation. Procedural automation follows predefined rules or workflows, whereas intelligent automation introduces data-driven inference, prediction, diagnosis, recommendation, or action. AI-enabled DevOps therefore extends conventional automation by enabling systems to interpret operational evidence and adapt decisions to context.

2.2 Artificial Intelligence and AIOps in DevOps

AI techniques used in DevOps include classical machine learning, deep learning, anomaly-oriented or unsupervised approaches, optimization and reinforcement learning, natural-language and representation-learning methods,

Generative AI and LLMs, and emerging agentic or multi-agent approaches. These techniques are applied to tasks such as anomaly detection, failure prediction, test optimization, resource management, root-cause analysis, incident triage, remediation support, and operational decision making.

AIOps is treated in this review as an application paradigm rather than a distinct algorithmic category. The term describes the application of analytics and AI to IT operations and failure management, while the underlying methods may belong to several AI families [2]. This distinction is important for avoiding the conflation of an operational domain with a specific technical method.

2.3 Generative AI, LLMs, and Agentic DevOps

Generative AI and LLMs broaden AI-for-DevOps by enabling reasoning over unstructured operational artifacts such as logs, tickets, runbooks, documentation, code, and incident narratives. Examples include recommending root causes and mitigation steps for cloud incidents [3] and broader applications of LLMs in software engineering [4].

Agentic systems extend LLM-based reasoning by integrating planning, tool use, execution, feedback, and multi-agent coordination. In a DevOps setting, this may enable systems not only to explain or recommend actions but also to invoke tools and carry out bounded operational steps. Because operational actions can affect availability, cost, security, and data integrity, agentic DevOps requires explicit permissions, evidence verification, auditability, rollback, human oversight, and safe escalation mechanisms [5].

For synthesis purposes, this review uses the functional lens Assist → Predict → Diagnose → Recommend → Act. It is a review-level conceptual lens for comparing forms of automation and should not be interpreted as a universally accepted external maturity model.

2.4 Scope Boundary: AI-for-DevOps versus MLOps and DevOps-for-AI

The review boundary is defined by the direction of intervention. AI-for-DevOps refers to the use of AI to improve DevOps processes, for example by automating testing, monitoring, diagnosis, incident handling, resource optimization, or operational decision support. DevOps-for-AI and MLOps instead use DevOps principles and infrastructure to manage the lifecycle of AI/ML systems [6].

Because some studies combine both directions, borderline cases were assessed according to the primary research objective and the operational process being improved. Studies focused exclusively on the development, deployment, or governance of AI models were excluded unless AI was also being used to improve a DevOps activity.

2.5 Conceptual Framework of the Review

The conceptual framework links the five research questions as follows: Research Evolution (RQ1) → DevOps Stage/Activity (RQ2) → AI Method/Technology (RQ3) → Benefits and Evidence Strength (RQ4) → Challenges and Limitations (RQ5) → Research Gaps and Future Directions.

This structure separates what is being automated, how AI is being applied, what benefits are reported, and how strongly those benefits are supported. It also supports a broader synthesis of technological evolution from model-centric prediction toward pipeline-integrated analytics, reasoning systems, and emerging agentic systems. This trajectory is an interpretive synthesis of the reviewed literature rather than a causal or universally sequential maturity model.

3. Research Methodology

3.1 Review Protocol and Research Questions

This SLR followed established systematic review guidance for software engineering, using PICOC to structure the scope and search process and PRISMA 2020 to report study selection. The review scope was limited to AI-for-DevOps research published from 2016 to 2026. The final database search was conducted on 7 September 2026.

The five research questions were defined to cover temporal trends (RQ1), DevOps stages and activities (RQ2), AI methods and technologies (RQ3), benefits and impacts (RQ4), and challenges, limitations, and future opportunities (RQ5). The synthesis combines descriptive quantitative analysis with qualitative thematic synthesis; no statistical meta-analysis was performed.

3.2 PICOC and Search Strategy

Table 1. PICOC Framework Used in the Review

PICOC Element	Operational Definition
Population	DevOps practices, processes, pipelines, CI/CD, and software delivery.
Intervention	Artificial Intelligence, Machine Learning, Deep Learning, Generative AI, Large Language Models, and AIOps.
Comparison	Traditional or non-AI DevOps approaches, where available; comparison was not mandatory for inclusion.
Outcomes	Automation, efficiency, reliability, software quality, deployment performance, incident resolution, and related operational outcomes.
Context	Software engineering, software development, and software operations.

The final review used operational search logs preserved for the three databases rather than retroactively presenting preliminary Boolean strings as exact executed queries. ScienceDirect searches comprised DevOps + AI (48), CI/CD + AI (28), continuous integration + AI (50), continuous delivery + AI (11), continuous deployment + AI (12), software delivery + AI (3), and AIOps (33), totaling 185 records. The corresponding IEEE Xplore searches returned 179, 129, 159, 38, 53, 23, and 184 records, totaling 765.

ACM Digital Library returned 182 records; two proceedings/container records were removed before screening, leaving 180 article records. The exact platform-specific ACM textual query syntax was not preserved in the available review records. Preliminary Boolean strings existed during protocol development but were not treated as the exact final executed query. Scopus Preview was considered during protocol development but was not part of the final primary database set.

3.3 Study Selection and PRISMA Process

The searches identified 1,132 records: 185 from ScienceDirect, 765 from IEEE Xplore, and 180 ACM article records after removing two container/proceedings records. Thus, 1,130 records entered deduplication. A total of 317 duplicate occurrences were removed, leaving 813 unique records for title and abstract screening.

Title and abstract screening excluded 431 records, leaving 382 reports sought for retrieval. Retrieval was attempted for all 382 reports through the review workflow. Seventy-three full texts were obtained and assessed, whereas 309 reports were not retrieved. Thirteen of the 73 assessed reports were excluded at full-text stage, leaving 60 included primary studies. The final exclusion reasons were FT3=2, FT4=1, FT5=7, and FT6=3.

The final PRISMA accounting is therefore: 1,132 identified → 1,130 entering deduplication → 813 unique screened → 382 reports sought → 73 retrieved and assessed → 60 included. No pending full-text decisions remained.

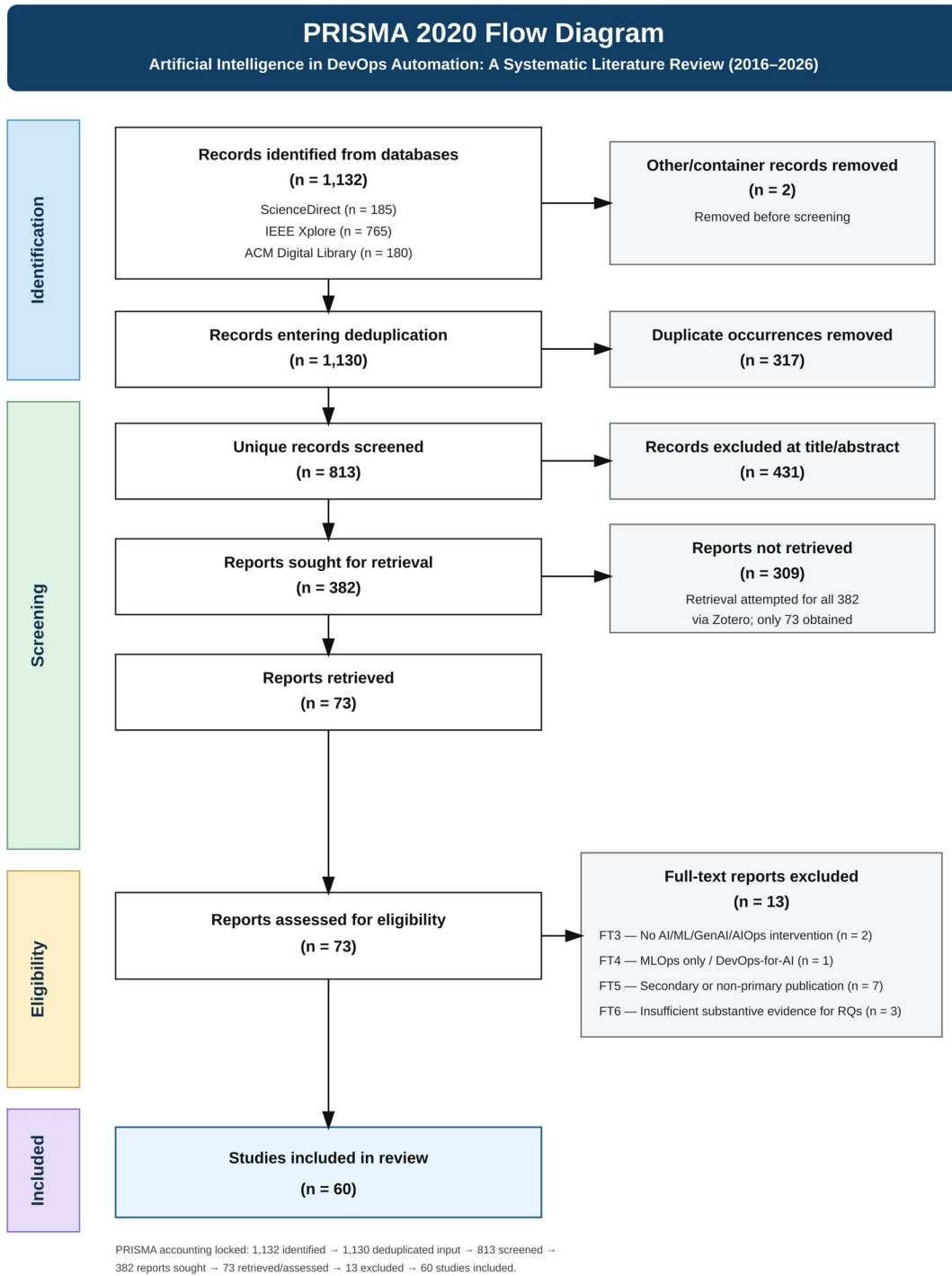


Figure 1. PRISMA 2020 flow diagram of the study-selection process.

3.4 Quality Assessment

All 60 included studies underwent quality assessment using five criteria: QA1, clarity of the research objective; QA2, clarity of the DevOps and AI relationship; QA3, adequacy of the research method or evaluation; QA4, evidence support for the reported results; and QA5, reporting of limitations, threats, or implications.

A study passed when it achieved at least 3.0 out of 5. All 60 included studies passed the threshold. QA PASS was used only as a minimum quality gate and was not interpreted as evidence that all studies had equal empirical strength.

3.5 Data Extraction

Data extraction was completed for all 60 studies using a 26-field schema: Paper ID, Authors, Year, Title, Source Database, Venue, Publication Type, Research Type, DevOps Stage, DevOps Activity, AI Category, AI Method/Model, Dataset/System, Research Objective, Evaluation Method, Metrics, Main Findings, Benefits, Challenges, Limitations, Future Work, RQ Mapping, QA Score, Industrial Context, Evidence Type, and Evidence Strength.

Main Findings, Benefits, Challenges, Limitations, and Future Work were retained as paper-derived evidence. DevOps Stage, AI Category, RQ Mapping, and Evidence Strength were treated as review-level coding. Evidence Strength was classified as Production, Industrial Empirical, Controlled Experimental, Benchmark, Simulation, or Conceptual.

3.6 Data Synthesis and Analysis

RQ1 was analyzed through annual descriptive counts and qualitative interpretation of technological trends. RQ2 used multi-label coding because a study could address multiple DevOps stages or activities; therefore, RQ2 percentages are not expected to sum to 100%. RQ3 used a seven-category taxonomy and a mutually exclusive primary category for quantitative distribution.

RQ4 synthesized seven benefit themes. Because reproducible corpus-wide B1–B7 frequencies were not available, benefit themes were not ranked statistically; representative evidence anchors were used instead. RQ5 synthesized fourteen challenge themes, eight research gaps (G1–G8), and five future research directions (FR1–FR5), again without inventing corpus-wide prevalence frequencies.

Interpretation was evidence-strength-aware. Benchmark or controlled experimental results were not treated as equivalent to production impact, higher predictive metrics were not assumed to imply lower MTTR, and emerging agentic systems were not characterized as broadly production-ready.

4. Results

4.1 Study Selection and Corpus Overview

The final corpus comprised 60 primary studies published between 2019 and 2026. No included study was published in 2016–2018. The corpus spans production deployments, industrial empirical studies, controlled experiments, benchmarks, simulations, and conceptual or applied architectural work.

The publication distribution is concentrated in the later years of the review period, with more than half of the included studies appearing in 2024–2026. This concentration is consistent with the rapid expansion of LLM-based and emerging agentic approaches in recent years.

4.2 RQ1 — Research Trends from 2016 to 2026

Table 2. Annual Distribution of Included Studies

Year	Studies
2016	0
2017	0
2018	0
2019	3
2020	7
2021	4
2022	7
2023	7
2024	12
2025	8
2026	12

The final annual distribution is 2019=3, 2020=7, 2021=4, 2022=7, 2023=7, 2024=12, 2025=8, and 2026=12. Study [7] is assigned to 2026 based on its final bibliographic publication in IEEE Internet of Things Journal, while the 2025 date corresponds to its earlier online/DOI publication record. This correction does not change the total corpus of 60 studies.

Across the period, the literature evolves from predominantly predictive AI toward diagnostic AIOps, followed by Generative AI and emerging agentic approaches. This result describes the observable research trajectory; deeper architectural interpretation is discussed in Section 5.

4.3 RQ2 — AI Applications Across DevOps Stages and Activities

Table 3. Distribution of AI Applications Across DevOps Stages and Activities

DevOps Stage / Activity	Studies	Percentage
Monitor / Observability	35	58.3%
Operate / Operations	31	51.7%
Incident / RCA / Remediation	17	28.3%
Test	16	26.7%
Build / CI	10	16.7%
Deploy / Release	9	15.0%
Develop / Code / IaC	3	5.0%
AIOps Model Governance	3	5.0%
Human / Cross-cutting	1	1.7%

RQ2 uses multi-label coding, so percentages do not sum to 100%. The largest concentrations occur in monitoring/observability and operations, followed by incident, root-cause analysis, and remediation. Testing, build/CI, and deployment also receive substantial attention, whereas development/code/IaC and explicitly human-centered or governance-oriented studies are less frequent.

4.4 RQ3 — AI Methods and Technologies

Table 4. Primary AI Category Distribution

Primary AI Category	Studies	Percentage
Classical Machine Learning	27	45.0%
Deep Learning	8	13.3%
Generative AI / LLM	7	11.7%
Unsupervised / Anomaly AI	6	10.0%
Agentic / Multi-Agent AI	5	8.3%
NLP / Representation Learning	4	6.7%
Optimization & Reinforcement Learning	3	5.0%

Classical Machine Learning remains the dominant primary category. At the same time, the corpus includes a growing methodological range spanning Deep Learning, anomaly-oriented techniques, NLP and representation learning, optimization and reinforcement learning, Generative AI/LLM, and emerging Agentic/Multi-Agent systems. AIOps is not counted as a separate algorithmic category.

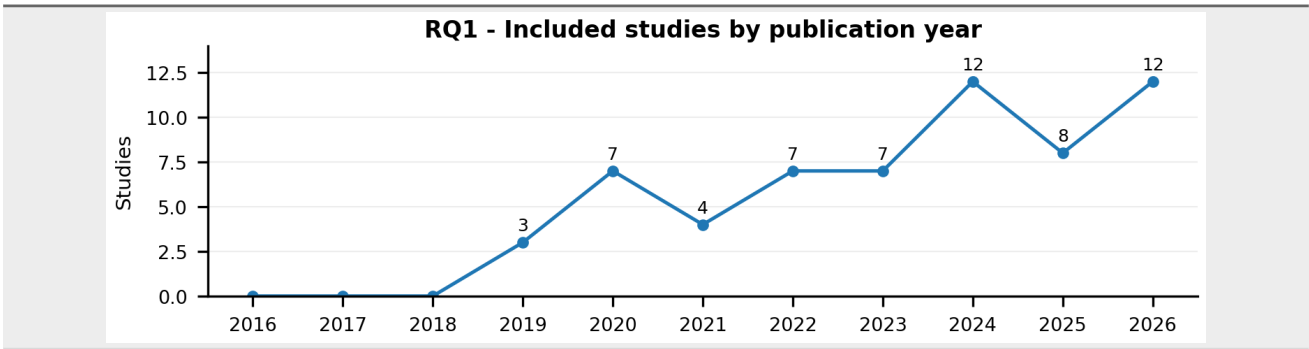


Figure 2. RQ1 publication trend.

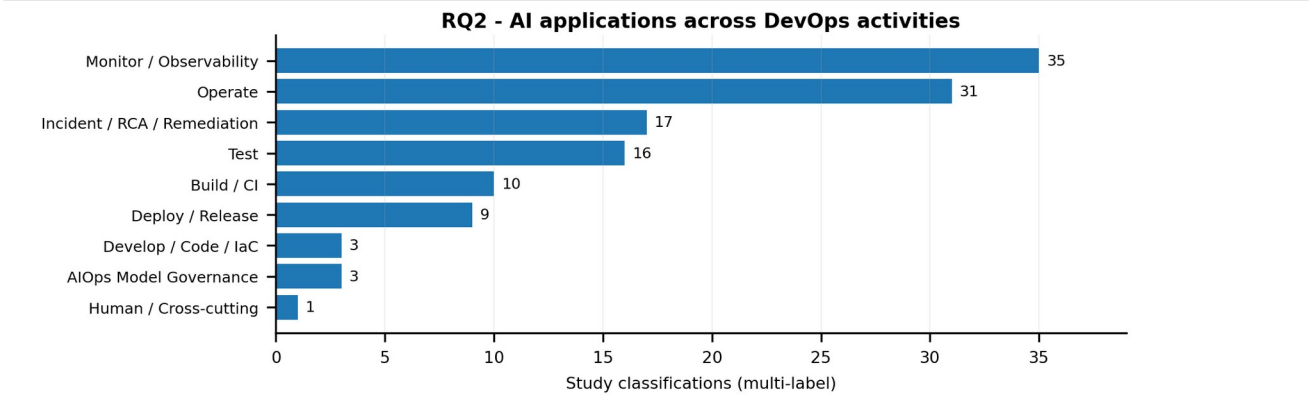


Figure 3. RQ2 DevOps activity distribution (multi-label).

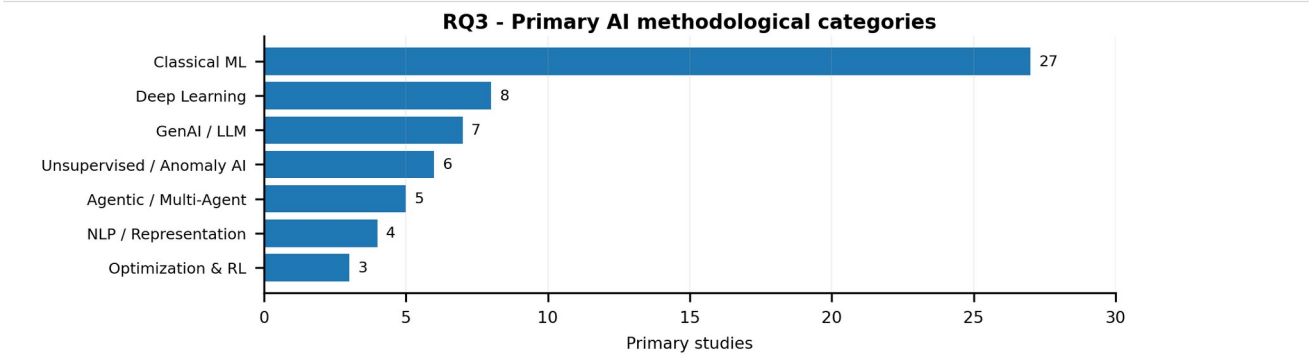


Figure 4. RQ3 primary AI category distribution.

4.5 RQ4 — Benefits and Impacts

The qualitative synthesis identified seven benefit themes: B1 Speed & Efficiency; B2 Accuracy & Detection Quality; B3 Reliability & Availability; B4 Cost & Resource Optimization; B5 Automation & Toil Reduction; B6 Security Improvement; and B7 Human Decision Support & Productivity.

These themes summarize the types of benefits reported in the corpus. They are not corpus-wide prevalence rankings because reproducible frequencies across all 60 studies were not available. Representative evidence anchors include production studies such as [8] (Predictive Test Selection), [9] (Alibaba node-failure prediction), and [10] (DeCaf); industrial empirical studies such as [11] (GPT-4 root-cause analysis) and [12] (LLM-based log prioritization); and benchmark evidence such as [13] (graph-augmented multi-agent RCA).

4.6 RQ5 — Challenges, Limitations, and Future Opportunities

Fourteen recurring challenge themes were identified: Data Quality & Availability; Class Imbalance; Data Leakage & Unrealistic Evaluation; Concept Drift & Model Decay; Limited Generalizability; Benchmark-to-Production Gap; Explainability & Trust; LLM Hallucination & Reliability; Cost/Latency/Resource Consumption; Integration Complexity; Safety of Autonomous Remediation; Human-in-the-Loop; Security & Privacy; and Evaluation Metric Inconsistency.

These challenges were synthesized into eight research gaps: G1 Production-Scale Validation; G2 Cross-System Generalization; G3 Concept Drift & Long-Term Maintenance; G4 Explainability, Trust & Reliability; G5 Autonomous-Remediation Safety; G6 LLM/Agent Cost & Latency; G7 Human-AI Collaboration & Adoption; and G8 Unified ML + DevOps Operational Evaluation.

The gaps were further consolidated into five future research directions: FR1 Production-Validated AI-for-DevOps; FR2 Adaptive and Generalizable AIOps; FR3 Trustworthy and Explainable LLM-AIOps; FR4 Safe and Cost-Efficient Agentic DevOps; and FR5 Human-Centered Autonomous DevOps. No corpus-wide challenge frequencies are reported.

For completeness, Table 5 summarizes all 60 included primary studies together with the coding used in the synthesis.

Table 5. Overview of the 60 Included Primary Studies

Note: PS identifiers denote the primary-study IDs used during study selection and data extraction; bracketed numbers indicate the corresponding IEEE references.

PS ID	Citation	Year	Title	DevOps Stage	AI Category	Evidence Strength
PS001	[14]	2025	Research on Big Data Analysis and Anomaly Detection Algorithms in Intelligent Operations and Maintenance	Monitor; Operate	Deep Learning	Controlled Experimental
PS002	[15]	2026	Context-Aware Predictive Optimization of Non-Functional Requirement Conflicts in DevOps Using Machine Learning	Monitor; Operate; runtime optimization	Optimization & RL / Classical ML	Controlled Experimental
PS003	[16]	2025	Anomaly Detection Algorithms for Real-Time Log Data Analysis at Scale	Monitor; Operate	Unsupervised / Anomaly AI	Industrial Empirical
PS004	[17]	2019	AIOps for a Cloud Object Storage Service	Monitor; Operate; Incident/RCA	Classical ML	Production
PS005	[18]	2025	AI-Augmented DevSecOps Pipelines for Secure and Scalable Service-Oriented Architectures in Cloud-Native Systems	Build; Test; Deploy; Monitor; Operate	Deep Learning	Controlled Experimental
PS006	[19]	2023	Scalable and Accurate Test Case Prioritization in Continuous Integration Contexts	Build/CI; Test	Classical ML	Benchmark
PS007	[20]	2026	Predicting Intermittent Job Failure Categories for Diagnosis Using Few-Shot Fine-Tuned Language Models	Build/CI; Incident diagnosis	Generative AI / LLM*	Industrial Empirical

PS ID	Citation	Year	Title	DevOps Stage	AI Category	Evidence Strength
PS008	[21]	2024	Interaction Prediction and Anomaly Detection in a Microservices-based Telecommunication Platform	Monitor; Operate; Incident	Unsupervised / Anomaly AI	Industrial Empirical
PS009	[22]	2024	Commit Artifact Preserving Build Prediction	Build / CI	NLP / Representation Learning	Benchmark
PS010	[23]	2022	Scalability Testing Automation using Multivariate Characterization and Detection of Software Performance Antipatterns	Test; CI/CD; performance monitoring	Classical ML	Industrial Empirical
PS011	[24]	2024	An AIOps Approach to Data Cloud Based on Large Language Models	Monitor; Operate; Incident; Automation	Generative AI / LLM	Conceptual / Applied
PS012	[10]	2020	DeCaf: Diagnosing and Triaging Performance Issues in Large-Scale Cloud Services	Monitor; Operate; Incident/RCA	Classical Machine Learning	Production
PS013	[25]	2023	On-Premise AIOps Infrastructure for a Software Editor SME: An Experience Report	Monitor; Operate; Incident Management	Classical Machine Learning	Production
PS014	[26]	2022	KubeKlone: A Digital Twin for Simulating Edge and Cloud Microservices	Operate; Resource Management	Classical Machine Learning	Simulation
PS015	[27]	2021	Automated User Experience Testing through Multi-Dimensional Performance Impact Analysis	Test; Build/CI	Classical Machine Learning	Controlled Experimental
PS016	[28]	2023	Security in DevSecOps: Applying Tools and Machine Learning to Verification and Monitoring Steps	Develop/IaC; Test; Deploy; Monitor	NLP / Representation Learning	Industrial Empirical
PS017	[29]	2020	Towards Intelligent Incident Management: Why We Need It and How We Make It	Monitor; Operate; Incident Management	NLP / Representation Learning	Industrial Empirical
PS018	[30]	2022	Characterizing and Triaging Change Points	Test; Monitor / Performance Engineering	Classical Machine Learning	Benchmark / Controlled Experimental
PS019	[31]	2025	Predictive Resource Management in the Computing Continuum: Transfer Learning from Virtual Machines to Containers	Operate; Resource Management	Deep Learning	Benchmark
PS020	[32]	2026	Leveraging Time Series Foundation Models to Detect Performance Anomalies in Software Systems	Monitor; Operate	Generative AI / LLM / Foundation Models	Benchmark
PS021	[33]	2024	An Architecture for Model-Based and Intelligent Automation in DevOps	Develop; Test; Deploy; Monitor; Operate	Classical Machine Learning*	Industrial Empirical
PS022	[34]	2025	Integrating Time Series Anomaly Detection Into DevOps Workflows	Test; Monitor / Observability	Unsupervised / Anomaly AI	Industrial Empirical
PS023	[35]	2019	Monitorless: Predicting Performance Degradation in Cloud Applications with Machine Learning	Test; Deploy; Monitor; Operate	Classical Machine Learning	Controlled Experimental
PS024	[36]	2022	Utilizing Persistence for Post Facto Suppression of Invalid Anomalies Using System Logs	Monitor; Operate	Unsupervised / Anomaly AI	Controlled Experimental
PS025	[37]	2026	Enhancing DevOps Continuous Monitoring Phase: Hybrid Intrusion Detection and Ensemble Learning System (HIDELS)	Test; Deploy; Monitor	Classical Machine Learning	Controlled Experimental
PS026	[38]	2026	“Still in the Loop”: Coping with Technostress in DevOps Teams and the Impact of GenAI	Cross-cutting / Human-centric	Generative AI / LLM	Industrial Empirical
PS027	[39]	2024	Security Anomaly Detection in Enterprise GitHub	Develop; Monitor; DevSecOps	Unsupervised / Anomaly AI	Industrial Empirical

PS ID	Citation	Year	Title	DevOps Stage	AI Category	Evidence Strength
PS028	[40]	2022	Towards a Consistent Interpretation of AIOps Models	Operate; AIOps Model Governance	Classical Machine Learning	Benchmark
PS029	[41]	2024	AIOps and Sustainability: Transforming Data Centers for a Greener Future	Monitor; Operate	Classical Machine Learning	Industrial Empirical
PS030	[42]	2024	Machine Learning-based Test Case Prioritization using Hyperparameter Optimization	Build / CI; Test	Optimization & Reinforcement Learning**	Benchmark
PS031	[43]	2022	An Intelligent Framework for Timely, Accurate, and Comprehensive Cloud Incident Detection	Monitor; Operate; Incident Management	Classical Machine Learning	Production
PS032	[44]	2023	Proactive Auto-Scaling Approach of Production Applications Using an Ensemble Model	Monitor; Operate; Resource Management	Classical Machine Learning	Industrial Empirical
PS033	[45]	2024	On the Model Update Strategies for Supervised Learning in AIOps Solutions	Operate; AIOps Model Governance / Maintenance	Classical Machine Learning	Benchmark
PS034	[46]	2021	Continuous Test Suite Failure Prediction	Build / CI; Test	Classical Machine Learning	Benchmark
PS035	[47]	2022	Within-Project Defect Prediction of Infrastructure-as-Code Using Product and Process Metrics	Develop / IaC; Test / QA	Classical Machine Learning	Benchmark
PS036	[48]	2026	Breaking the Observability Tax: Dynamic Resolution Anomaly Detection via Topology-Aware Active LLM Agents	Monitor; Operate; Incident Diagnosis	Agentic / Multi-Agent AI	Simulation
PS037	[49]	2026	Data-Driven Test Case Prioritization (DD-TCP): A Machine Learning Framework for Intelligent Software Quality Assurance	Build / CI; Test	Classical Machine Learning	Benchmark
PS038	[50]	2023	Knowledge-based Intelligent System for IT Incident DevOps	Operate; Incident Management	NLP / Representation Learning	Industrial Empirical
PS039	[51]	2020	Towards AIOps in Edge Computing Environments	Monitor; Operate; Remediation	Classical Machine Learning	Controlled Experimental
PS040	[52]	2024	Advancing Software Security and Reliability in Cloud Platforms through AI-based Anomaly Detection	Build; Test; Deploy; Monitor	Deep Learning	Controlled Experimental
PS041	[53]	2025	Cross-System Categorization of Abnormal Traces in Microservice-Based Systems via Meta-Learning	Monitor; Operate; Incident/RCA	Deep Learning	Benchmark
PS042	[12]	2026	LLM-based AIOps via Log Prioritization in Air-Gapped Systems	Monitor; Operate; Incident Diagnosis	Generative AI / LLM	Industrial Empirical
PS043	[54]	2021	KPI Anomaly Detection Method for Data Center AIOps Based on GRU-GAN	Monitor; Operate	Deep Learning	Benchmark
PS044	[55]	2023	A Case Study of AIOps in Large Enterprises Using Predictive Analytics for IT Operations	Monitor; Operate	Classical Machine Learning	Industrial Empirical
PS045	[7]	2026	ARM: Autonomous Remediation & Management with LLM Agents for Intent-Driven Control	Monitor; Operate; Incident/RCA; Remediation	Agentic / Multi-Agent AI	Controlled Experimental
PS046	[56]	2026	LLM Agents for AIOps in Kubernetes: An Industrial Experience Report with Red Hat OpenShift	Monitor; Operate; Capacity Planning; Remediation	Agentic / Multi-Agent AI	Industrial Empirical
PS047	[57]	2024	Commit Artifact Preserving Build Prediction	Build / CI	NLP / Representation Learning†	Benchmark

PS ID	Citation	Year	Title	DevOps Stage	AI Category	Evidence Strength
PS048	[58]	2024	LogExpert: Log-based Recommended Resolutions Generation using Large Language Model	Monitor; Operate; Incident/Remediation	Generative AI / LLM	Benchmark
PS049	[13]	2026	Graph-Augmented Multi-Agent Robust Root Cause Analysis in AIOps	Monitor; Operate; Incident/RCA	Agentic / Multi-Agent AI	Benchmark
PS050	[59]	2026	DualLane: Fast and Reliable LLM Agents for Interactive AIOps via Dual-Path Planning	Operate; Incident/Support; Resource Operations	Agentic / Multi-Agent AI	Production
PS051	[60]	2020	An Experimental Evaluation of Imbalanced Learning and Time-Series Validation in the Context of CI/CD Prediction	Build / CI; Deploy / CD	Classical Machine Learning	Benchmark
PS052	[61]	2025	Fault Prediction in Large-Scale Cluster Environments Based on Improved TCN and DenseNet	Monitor; Operate	Deep Learning	Controlled Experimental
PS053	[62]	2024	On the Effectiveness of Feature Selection Techniques in the Context of ML-Based Regression Test Prioritization	Build / CI; Test	Classical Machine Learning	Benchmark
PS054	[63]	2020	An Anomaly Detection Algorithm for Microservice Architecture Based on Robust Principal Component Analysis	Monitor; Operate; Incident/RCA	Unsupervised / Anomaly AI	Benchmark
PS055	[64]	2020	Env2Vec: Accelerating VNF Testing with Deep Learning	Build; Test; Pre-deployment	Deep Learning	Industrial Empirical
PS056	[8]	2019	Predictive Test Selection	Build / CI; Test	Classical Machine Learning	Production
PS057	[9]	2020	Predicting Node Failures in an Ultra-Large-Scale Cloud Computing Platform: An AIOps Solution	Monitor; Operate	Classical Machine Learning	Production
PS058	[65]	2021	An Empirical Study of the Impact of Data Splitting Decisions on the Performance of AIOps Solutions	Operate; AIOps Model Governance / Maintenance	Classical Machine Learning	Benchmark
PS059	[66]	2025	PA-Rank: A GAN and Reinforcement Learning Powered Framework for Multi-Metric Anomaly Detection and Causal Diagnosis	Monitor; Operate; Incident/RCA	Optimization & Reinforcement Learning	Benchmark
PS060	[11]	2024	Automated Root Causing of Cloud Incidents using In-Context Learning with GPT-4	Operate; Incident/RCA	Generative AI / LLM	Industrial Empirical

5. Discussion

5.1 Evolution of AI-Enabled DevOps

The results indicate overlapping phases rather than simple replacement of older approaches. Earlier work is primarily model-centric, using prediction, classification, anomaly detection, and diagnostic analytics. As these methods become embedded in operational workflows, the unit of automation expands from isolated models toward pipelines that collect evidence, trigger analysis, and support operational decisions.

LLMs extend this architecture by enabling reasoning over heterogeneous and unstructured operational evidence. This supports a transition from Model → Pipeline → Reasoning System. Emerging agentic systems add planning, tools, verification, and coordinated execution, extending the trajectory toward Agentic Systems. In parallel, the functional depth of automation can be interpreted as Assist → Predict → Diagnose → Recommend → Act.

Newer approaches do not automatically replace established machine-learning methods. Classical and anomaly-oriented techniques remain useful where their performance, efficiency, interpretability, and operational

requirements are well understood. Generative and agentic systems broaden the range of tasks that can be addressed but also introduce new reliability, cost, and safety concerns.

Overall, the corpus supports a broader evolution from Predictive AI → Diagnostic AIOps → Generative AI → Agentic AI. However, technical capability is advancing faster than sustained evidence of production benefit, especially for LLM-based and agentic approaches. A longer-term direction can therefore be expressed as Predictive → Diagnostic → Generative → Agentic → Trustworthy Autonomous DevOps, with the final stage understood as a research direction rather than an established maturity level.

5.2 Operational Concentration and Automation Maturity

The RQ2 distribution reveals a strong shift-right concentration: monitoring and observability, operations, and incident/RCA/remediation dominate the corpus. These stages generate continuous telemetry and recurring operational problems that are naturally suited to data-driven analysis, uncertainty reduction, and repetitive decision support.

Lower counts in development, build, and deployment do not imply that AI is irrelevant to earlier lifecycle stages. Rather, they reflect the scope and framing of the reviewed literature, which emphasizes operational data and failure management.

The two-dimensional view of maturity combines breadth of operational involvement—Monitoring → Operations → Diagnosis → Remediation—with depth of automation—Assist → Predict → Diagnose → Recommend → Act. As systems move toward Act, the consequences of errors become more significant. Evidence verification, permissions, human approval for high-impact actions, safe execution, auditability, rollback, and escalation therefore become increasingly important.

Automation maturity should thus be judged not only by model sophistication but also by operational responsibility and evidence strength.

5.3 Evidence Maturity and the Benchmark-to-Production Gap

The six evidence-strength categories provide an important lens for interpreting RQ4. Production evidence offers the strongest direct indication that an approach can operate under real organizational and system constraints. Industrial empirical studies provide realistic evidence but may still be limited in duration, scope, or operational integration. Controlled experiments and benchmarks are essential for technical comparison, while simulation and conceptual work primarily demonstrate feasibility.

Production examples such as [8], [9], and [10] demonstrate direct operational deployment. [11] and [12] provide industrial empirical evidence, while [13] demonstrates benchmark-level technical capability. These evidence types should not be treated as interchangeable.

Production environments introduce system evolution, concept drift, noisy telemetry, tool integration, security constraints, workload variability, latency, cost, and consequences of incorrect actions. For this reason, strong benchmark accuracy or F1 scores should not be interpreted as proof of reduced MTTR or improved availability.

More mature evaluation should connect algorithmic performance with operational outcomes such as incident-resolution time, false-action rates, time to detection or diagnosis, MTTR, availability, engineering effort, human intervention, cost, latency, and safety. Evidence maturity therefore requires both technical performance and demonstrated operational effectiveness.

5.4 Research Gaps

G1 Production-Scale Validation. More studies are needed that evaluate AI-for-DevOps under sustained, realistic production conditions and report operational outcomes rather than only offline model metrics.

G2 Cross-System Generalization. Models frequently depend on system-specific telemetry, architecture, or incident patterns. Research should evaluate transfer across systems, organizations, domains, and infrastructure configurations.

G3 Concept Drift and Long-Term Maintenance. DevOps environments continuously evolve. Future work should address drift detection, continual adaptation, retraining strategies, and monitoring of the AI system itself.

G4 Explainability, Trust, and Reliability. Operational decisions require grounded evidence, uncertainty estimation, traceability, and mechanisms to reduce hallucinations and unsupported recommendations.

G5 Autonomous-Remediation Safety. Action-capable systems require bounded autonomy, verification, policy enforcement, authorization, rollback, auditability, and escalation. The central question is under what conditions an AI system should be allowed to act.

G6 LLM/Agent Cost and Latency. Multi-step reasoning, tool calls, long contexts, and multi-agent interaction can introduce significant computational and response-time overhead. Capability must be balanced against operational efficiency.

G7 Human-AI Collaboration and Adoption. Research should examine supervision, approval, trust, accountability, skills, organizational adoption, and how AI changes rather than simply replaces operational work.

G8 Unified ML and DevOps Operational Evaluation. Evaluation frameworks should connect model-level metrics with DevOps-level outcomes, measuring both prediction or reasoning quality and the downstream operational effect.

As a synthesis-level prioritization rather than statistical prevalence, G1, G5, and G8 emerge as the highest-priority gaps; G2, G3, and G4 are also high-priority; and G6 and G7 are emerging high-priority areas.

5.5 Future Research Agenda

FR1 Production-Validated AI-for-DevOps. Longitudinal production studies should evaluate reliability, operational outcomes, maintenance burden, cost, and safety over time.

FR2 Adaptive and Generalizable AIOps. Future systems should incorporate drift awareness, continual learning, cross-system transfer, domain adaptation, and monitoring of their own performance.

FR3 Trustworthy and Explainable LLM-AIOps. Research should prioritize grounding, hallucination mitigation, uncertainty, explanations, traceability, consistency, verification, and protection of sensitive operational information.

FR4 Safe and Cost-Efficient Agentic DevOps. Agentic systems should use bounded tool access, explicit permissions, verification, policy constraints, sandboxing, rollback, audit logs, escalation, and cost/latency controls. The goal is appropriate autonomy under explicit safety constraints rather than maximum autonomy.

FR5 Human-Centered Autonomous DevOps. Future systems should support graduated patterns such as AI recommends → Human approves; AI acts within bounded policy → Human supervises; and AI acts autonomously on low-risk tasks → Human handles escalation. Trust, interface design, workload, accountability, adoption, and skills remain central.

5.6 Toward Trustworthy Autonomous DevOps

A trustworthy autonomous DevOps system must be evaluated across multiple dimensions: technical capability, reliability, generalizability, explainability and evidence grounding, safety, security and privacy, economic feasibility, and human governance.

A practical autonomy progression is Human-led → AI-assisted → AI-recommended → Human-supervised automation → Bounded autonomous operations. Maturity therefore cannot be reduced to model sophistication alone.

The long-term trajectory suggested by this review is Predictive → Diagnostic → Generative → Agentic → Trustworthy Autonomous DevOps. The central research question consequently changes from “Can AI automate this DevOps task?” to “Under what evidence, constraints, and governance can AI perform this task safely and reliably in production?” The key challenge is closing the gap between technical possibility and trustworthy operational adoption.

6. Threats to Validity

6.1 Construct Validity

Construct validity concerns whether the operational definitions and classifications used in this review adequately represent AI for DevOps automation. The review deliberately distinguishes AI-for-DevOps from studies focusing

exclusively on MLOps or DevOps-for-AI. However, these areas may overlap in practice, making some borderline studies sensitive to reviewer interpretation.

Classification also introduces potential subjectivity. For RQ2, studies were mapped to DevOps stages and activities using a multi-label approach because individual studies may address multiple stages, particularly monitoring, operations, incident management, root-cause analysis, and remediation. Consequently, RQ2 percentages are not mutually exclusive and should not be expected to sum to 100%.

For RQ3, AI approaches were classified into Classical Machine Learning, Deep Learning, Unsupervised/Anomaly AI, Optimization and Reinforcement Learning, NLP/Representation Learning, Generative AI/LLM, and Agentic/Multi-Agent AI. AIOps was treated as an application paradigm rather than an algorithmic category. A mutually exclusive primary category was used for quantitative reporting, which improves interpretability but may simplify studies combining multiple AI techniques.

These risks were mitigated through explicit scope boundaries, predefined research questions, consistent classification rules, and a structured extraction schema. Nevertheless, DevOps Stage, AI Category, RQ Mapping, and Evidence Strength remain review-level analytical classifications rather than necessarily author-defined labels.

6.2 Internal Validity

The principal threat to internal validity is full-text retrieval bias. The search identified 1,132 records. After two records were removed before screening and 317 duplicate occurrences were eliminated, 813 unique records underwent title and abstract screening. Of these, 431 were excluded and 382 reports were sought for retrieval.

Retrieval was attempted for all 382 reports, but only 73 full texts were obtained, while 309 reports were not retrieved. Eligibility assessment was therefore restricted to the 73 available reports. This represents a substantial limitation because potentially relevant studies may be present among the unavailable reports. Accordingly, the 60 included studies should not be interpreted as representing all potentially eligible studies among the 382 reports sought for retrieval.

Of the 73 retrieved reports, 13 were excluded during full-text assessment, resulting in 60 included studies. All 60 studies passed the predefined quality-assessment threshold and underwent complete data extraction using the 26-field extraction schema. However, QA PASS indicates satisfaction of the minimum quality threshold and does not imply equal evidential strength. The included corpus spans production, industrial empirical, controlled experimental, benchmark, simulation, and conceptual evidence.

Reviewer judgment during screening, extraction, and thematic synthesis also introduces potential classification bias. The use of explicit eligibility criteria, structured extraction fields, consistent coding, and transparent PRISMA accounting was intended to reduce, but cannot completely eliminate, this threat.

6.3 External Validity

External validity is limited by database coverage and the characteristics of the included evidence. The search covered ScienceDirect, IEEE Xplore, and ACM Digital Library for the period 2016–2026. Although these sources provide substantial coverage of software engineering and computing research, the review does not claim exhaustive coverage of all AI-for-DevOps literature.

Relevant studies indexed exclusively in other databases, publisher platforms, institutional repositories, grey literature, or industrial technical reports may therefore be absent. This limitation is particularly relevant to rapidly developing areas such as LLM-based DevOps and agentic systems, where industrial developments may precede formal academic publication.

Generalization is further constrained by differences in datasets, software systems, infrastructure, organizational environments, and evaluation designs. Evidence obtained from benchmarks, simulations, controlled experiments, or conceptual studies cannot automatically be generalized to production environments. The synthesis therefore distinguishes evidence strength and avoids treating experimental performance as equivalent to demonstrated industrial impact.

6.4 Conclusion Validity

Conclusion validity concerns whether the conclusions are proportional to the available evidence. This review employs descriptive quantitative analysis and qualitative thematic synthesis, rather than statistical meta-analysis, because the included studies differ substantially in research design, datasets, metrics, contexts, and evidence strength.

For RQ4, seven benefit themes were identified, but reproducible corpus-wide frequencies across all 60 studies are not available. Therefore, the review does not infer prevalence rankings among the benefit themes. Visualizations based on representative evidence anchors indicate coverage among those anchors rather than frequency across the entire corpus.

Similarly, the fourteen challenge themes and eight research gaps synthesized for RQ5 do not have reproducible corpus-wide frequencies. Their prioritization should therefore be interpreted as evidence-informed synthesis rather than statistical prevalence.

The review further adopts conservative interpretation rules. Benchmark performance is not considered equivalent to production impact; improved predictive metrics do not necessarily imply improvements in operational measures such as MTTR; AI is not assumed to replace DevOps engineers; and emerging agentic AIOps approaches are not considered broadly production-ready without sufficient production evidence.

Overall, the findings should be interpreted as a systematic synthesis of the accessible and eligible literature captured by the review protocol, rather than an exhaustive or statistically representative account of all AI-for-DevOps research. Validity risks were mitigated through explicit scope definition, transparent PRISMA accounting, systematic screening, quality assessment, structured data extraction, evidence-strength-aware synthesis, and conservative reporting of quantitative claims.

7. Conclusion

This systematic literature review investigated the development, applications, methods, benefits, challenges, and future directions of Artificial Intelligence for DevOps automation between 2016 and 2026. The review specifically focused on AI-for-DevOps, in which AI techniques are used to improve or automate DevOps activities, while studies concerned exclusively with MLOps or DevOps-for-AI were outside the review scope.

The review identified 1,132 database records from ScienceDirect, IEEE Xplore, and ACM Digital Library. After removal, deduplication, title and abstract screening, full-text retrieval, and eligibility assessment, 60 primary studies were included in the final corpus. All included studies passed the predefined quality-assessment threshold and were analyzed using a structured 26-field data-extraction schema.

Regarding RQ1, the results show that research activity became increasingly concentrated in the later years of the review period. More than half of the included studies were published during 2024–2026. Across the corpus, the technological development of AI-for-DevOps reflects a progression from predominantly predictive and diagnostic approaches toward Generative AI, LLM-based systems, and emerging agentic or multi-agent approaches.

Regarding RQ2, AI applications were concentrated primarily in monitoring and observability, operations, and incident/root-cause/remediation activities. This indicates a strong operational orientation in the reviewed literature, while comparatively fewer studies addressed development, build, and deployment activities.

Regarding RQ3, Classical Machine Learning remained the dominant primary AI category, while the corpus also included Deep Learning, Unsupervised/Anomaly AI, Optimization and Reinforcement Learning, NLP/Representation Learning, Generative AI/LLM, and Agentic/Multi-Agent AI. AIOps was treated as an application paradigm rather than an algorithmic category. The coexistence of these approaches indicates methodological diversification rather than the replacement of established methods by newer technologies.

Regarding RQ4, the qualitative synthesis identified recurring benefits related to speed and efficiency, accuracy and detection quality, reliability and availability, cost and resource optimization, automation and toil reduction, security improvement, and human decision support and productivity. These themes should not be interpreted as corpus-wide prevalence rankings because reproducible frequencies across all 60 studies were not available. Moreover, the strength of evidence supporting reported benefits varied substantially across Production, Industrial Empirical, Controlled Experimental, Benchmark, Simulation, and Conceptual settings.

This distinction leads to one of the central conclusions of the review: technical capability is developing faster than demonstrated production benefit, particularly for LLM-based and agentic approaches. Strong benchmark or experimental performance provides valuable evidence of technical feasibility, but it should not be treated as equivalent to demonstrated operational impact.

Regarding RQ5, the review identified fourteen recurring challenge themes covering data quality, class imbalance, unrealistic evaluation, concept drift, generalizability, the benchmark-to-production gap, explainability, LLM reliability, cost and latency, integration complexity, autonomous-remediation safety, human oversight, security and privacy, and evaluation inconsistency.

These challenges were synthesized into eight research gaps: production-scale validation, cross-system generalization, concept drift and long-term maintenance, explainability/trust/reliability, autonomous-remediation safety, LLM/agent cost and latency, human–AI collaboration and adoption, and unified ML-plus-DevOps operational evaluation.

The eight gaps were further consolidated into five future research directions: Production-Validated AI-for-DevOps, Adaptive and Generalizable AIOps, Trustworthy and Explainable LLM-AIOps, Safe and Cost-Efficient Agentic DevOps, and Human-Centered Autonomous DevOps.

Taken together, the reviewed literature suggests a broader technological trajectory: Predictive → Diagnostic → Generative → Agentic → Trustworthy Autonomous DevOps.

The final stage should not be interpreted as an established level of industrial maturity. Rather, it represents a research direction in which increasing AI capability must be accompanied by stronger evidence of reliability, generalizability, safety, economic feasibility, explainability, governance, and appropriate human oversight.

The central challenge for future AI-enabled DevOps research is therefore shifting from demonstrating whether AI can perform an operational task toward determining under what evidence, constraints, and governance conditions AI can perform that task safely and reliably in production.

Future progress will depend not only on increasingly capable AI techniques, but on closing the gap between technical possibility and trustworthy operational adoption.

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